

1406 Amikacin Liposome Inhalation Suspension (ALIS) ALIS is a new formulation of the aminoglycoside amikacin, which allows for improved penetration in the lung with reduced toxicity. It is now approved for treatment of refractory MAC lung infection with persistent sputum positivity at 6 months while on appropriate background regimen. Typical dose is 590 mg (one vial once a day) for 6 months along with the standard three-drug regimen of macrolide, rifampin, and ethambutol. Dosage adjustments in patients with hepatic and renal dysfunction are not required. Half-life elimination typically occurs in ~5.9–9.5 h. Respiratory side effects such as bronchospasm, cough, dysphonia, and dyspnea are common. Monitoring for systemic aminoglycoside toxicity should be considered.

Imipenem Imipenem primarily inhibits cell-wall biosynthesis by binding to the penicillin-binding proteins. It is rapidly gaining importance for the treatment of *M. abscessus* with a meta-analysis showing improved outcomes with its inclusion in a multidrug regimen. It is dosed at 500 mg to 1 g twice to three times a day as part of a combination regimen for the treatment of *M. abscessus*. Half-life of imipenem is ~1 h, and because it is metabolized in the kidneys, dosing adjustment is needed with renal dysfunction. Adverse effects include anemia, thrombocytopenia, and liver dysfunction.

Cefoxitin Cefoxitin is a second-generation parenteral cephalosporin with activity against rapidly growing NTM, particularly *M. abscessus* and *M. chelonae*. Its mechanism of action against NTM is unknown but may involve inactivation of cell-wall synthesis enzymes. High doses are used for treatment of NTM: 200 mg/kg IV three or four times per day, with a maximal daily dose of 12 g. The half-life of cefoxitin is ~1 h, with primary renal clearance that requires adjustment in renal insufficiency. Adverse effects are uncommon but include gastrointestinal manifestations, rash, eosinophilia, fever, and neutropenia.

Newer Drugs Three newer classes of drugs—the oxazolidinones, the glycolcyclines, and the ketolides—are currently being evaluated for possible use in the treatment of NTM infections, especially those caused by *M. abscessus*. Approximately 50% of *M. abscessus* isolates have shown some degree of susceptibility in vitro to linezolid, an oxazolidinone. Tigecycline, which is a glycolcycline and a tetracycline derivative, and telithromycin, a ketolide, also appear to have in vitro activity against *M. abscessus*. These drugs, however, have not yet been clinically tested in patients.

In addition, some anti-TB drugs, including clofazimine and bedaquiline, are being evaluated as alternative agents for the treatment of refractory NTM infections. In particular, clofazimine appears to act synergistically in combination with amikacin, bedaquiline, or tigecycline. The exact role of these agents in the treatment of refractory NTM infections remains unclear. Suppressive therapy with periodic parenteral/oral drugs to limit disease progression and control symptoms may be an appropriate alternative to curative treatment.

CONCLUSION

Treatment of mycobacterial infections requires multiple-drug regimens that often exert significant side effects with the potential to limit tolerability. The prolonged duration of treatment has vastly improved results over those obtained in past decades, but drugs and regimens that will shorten treatment duration and limit adverse drug effects and interactions are needed.

FURTHER READING

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- DALEY CL et al: Treatment of nontuberculous mycobacterial pulmonary disease: An official ATS/ERS/ESCMID/IDSA clinical practice guideline. *Clin Infect Dis* 71:e1, 2020.
- NAHID P et al: Official American Thoracic Society/Centers for Disease Control and Prevention/Infectious Diseases Society of America clinical practice guidelines: Treatment of drug-susceptible tuberculosis. *Clin Infect Dis* 63:e147, 2016.

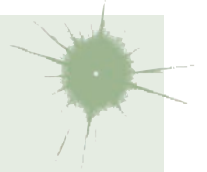
STERLING TR et al: Guidelines for the treatment of latent tuberculosis infection: Recommendations from the National Tuberculosis Controllers Association and CDC, 2020. *MMWR Recomm Rep* 69 (No. RR-1):1, 2020.

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Section 9 Spirochetal Diseases

182 Syphilis

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DEFINITION

Syphilis, a chronic systemic infection caused by *Treponema pallidum* subspecies *pallidum*, is usually sexually transmitted and is characterized by episodes of active disease interrupted by asymptomatic periods (latency). After an incubation period averaging 2–6 weeks, a primary lesion appears—often associated with regional lymphadenopathy—and then resolves without treatment. The secondary stage, with generalized mucosal and cutaneous lesions and generalized lymphadenopathy, also resolves spontaneously and is followed by a latent period of subclinical infection lasting years or decades. Central nervous system (CNS) invasion may occur early in infection, and CNS involvement may be symptomatic or asymptomatic. In the preantibiotic era, one-third of untreated patients developed tertiary syphilis, characterized by destructive mucocutaneous, skeletal, or parenchymal lesions; aortitis; or late CNS manifestations.

ETIOLOGY

The Spirochaetales include five genera that are pathogenic for humans and for a variety of other animals: *Leptospira* species (leptospirosis, **Chap. 184**); *Borrelia* and *Borrelia* species (relapsing fever and Lyme disease, respectively; **Chaps. 185 and 186**); *Brachyspira* species (gastrointestinal infections); and *Treponema* species (syphilis and the endemic treponematoses; **see also Chap. 183**). The *Treponema* subspecies include *T. pallidum* subsp. *pallidum* (venereal syphilis); *T. pallidum* subsp. *pertenue* (yaws); *T. pallidum* subsp. *endemicum* (endemic syphilis or bejel); and *T. carateum* (pinta). Historically, the subspecies were distinguished by the clinical syndromes they produce, but phylogenetic analyses of whole genome sequences from a relatively small number of strains (excluding *T. carateum*) yield the three named subspecies groupings. Whether these groupings represent geographical variation or true biological differences is unclear. The crossing of subspecies boundaries by some “molecular signatures” and the recent recognition of treponemes of the *endemicum* genotype in sexually acquired genital ulcers (chancres) and secondary rashes (**Chap. 183**) support the concept of a genetic and clinical “continuum” among strains and subspecies of the pathogenic treponemes.

T. pallidum subspecies are thin spiral organisms, with a cell body surrounded by a trilaminar cytoplasmic membrane, a delicate peptidoglycan layer, and a lipid-rich outer membrane. Endoflagella wind around the cell body in the periplasmic space and are responsible for motility.



Historically, *T. pallidum* could not be cultured in vitro, but long-term propagation of the Nichols strain of *T. pallidum* in complex medium with eukaryotic cells was recently reported. To date, the *pertenue* and *endemicum* subspecies have not been cultured. All *T. pallidum* subspecies have severely limited metabolic capabilities and are highly dependent on host-derived amino acids, carbohydrates, and lipids. Genetic analyses have revealed the existence of a 12-member

gene family (*tpr*) encoding outer-membrane antigens. One member, TprK, has discrete variable regions that undergo antigenic variation during infection, providing a mechanism for immune evasion.

The only known natural host for *T. pallidum* subsp. *pallidum* (referred to hereafter as *T. pallidum*) is the human. *T. pallidum* can infect many mammals, but only humans, higher apes, and a few laboratory animals develop syphilitic lesions. Rabbits are used to propagate *T. pallidum* and serve as the animal model that best reflects human disease and immunopathology.

TRANSMISSION AND EPIDEMIOLOGY

Nearly all cases of syphilis are acquired by sexual contact with infectious lesions (i.e., the chancre, mucous patch, skin rash, or condylomata lata; see Fig. A1-20). Less common modes of transmission include nonsexual skin contact, infection in utero, blood transfusion, and organ transplantation.

■ SYPHILIS IN THE UNITED STATES

Following the introduction of penicillin therapy in the 1940s, the number of cases of syphilis of all stages reported in the United States declined 95% to a low of 31,575 cases in 2000, with 5979 reported cases of primary and secondary (P&S) syphilis. (P&S cases are infectious and are a better indicator of disease activity than total syphilis cases.) Since 2000, total cases have increased 3.6-fold to 115,045, and the number of P&S cases has increased nearly sixfold, with 35,063 cases reported in 2018 (Fig. 182-1). Nationally, ~54% of these cases were in men who have sex with men (MSM), ~41.6% of whom are co-infected with HIV. From 2017 to 2018, P&S cases rose 11.7% among all men and 34.2% among women, with increases in all racial and ethnic groups and in all geographic regions of the United States. Because the incidence of congenital syphilis parallels that of infectious syphilis in women, the striking increase in early syphilis in women has resulted in a dramatic increase in congenital syphilis. In 2018, 1306 cases of congenital syphilis were reported, resulting in 78 stillbirths and 16 infant deaths. Since 2014, the number of reported cases in infants <1 year of age has increased nearly threefold. Data from 2018 show that nearly 12% of women with syphilis reported injecting drugs in the past 12 months and nearly 8% reported heroin use.

The populations at highest risk for acquiring syphilis have changed over time, with outbreaks among MSM in the pre-HIV era of the late 1970s and early 1980s, as well as at present. The dramatic recent increases in syphilis and other sexually transmitted infections in MSM may be due to unprotected sex between persons who are HIV concordant and to disinhibition facilitated by highly effective antiretroviral therapy (ART). Many MSM diagnosed with syphilis have had syphilis previously, and more frequent (every 3 months) screening for syphilis and other sexually transmitted infections is warranted

in such high-risk populations. Cases of P&S syphilis among African Americans increased 2.7-fold between 2000 and 2018, and the rate (28.1 per 100,000 population) remains higher than rates for other racial/ethnic groups.

Of individuals named as sexual contacts of persons with infectious syphilis, many will have developed manifestations of syphilis when they are first seen, and ~30% of asymptomatic contacts examined within 30 days of exposure actually have incubating infection and will later develop infectious syphilis if not treated. Thus, identification and treatment of all recently exposed sexual contacts continue to be important aspects of syphilis control.

■ GLOBAL SYPHILIS

Syphilis remains a significant health problem globally; the number of new infections is estimated at 6–8 million per year. The regions most affected include sub-Saharan Africa, South America, China, and Southeast Asia. The incidence rate for total syphilis in China continues to rise, and rates of P&S syphilis have increased dramatically among MSM in many European, Asian, and South American countries. Globally, through efforts by World Health Organization (WHO), progress has been made in the prevention of congenital syphilis, although there are still ~988,000 pregnant women with syphilis per year, with 661,000 cases of congenital syphilis including 200,000 stillbirths and infant deaths.

NATURAL COURSE AND PATHOGENESIS OF UNTREATED SYPHILIS

T. pallidum rapidly penetrates intact mucous membranes or microscopic abrasions in skin and, within a few hours, enters the lymphatics and blood to produce systemic infection and metastatic foci long before the appearance of a primary lesion. Blood from a patient with incubating or early syphilis is infectious. The generation time of *T. pallidum* during early disease is estimated to be ~30 h, and the incubation period of syphilis is inversely proportional to the number of organisms transmitted. The 50% infectious dose for intradermal inoculation in humans has been calculated to be 57 organisms, and the treponeme concentration generally reaches 10^7 /g of tissue before a clinical lesion appears. The median incubation period in humans (~21 days) suggests an average inoculum of 500–1000 infectious organisms for naturally acquired disease; the incubation period rarely exceeds 6 weeks.

The primary lesion appears at the site of inoculation, usually persists for 4–6 weeks, and then heals spontaneously. Histopathologic examination shows perivascular infiltration, chiefly by CD4+ and CD8+ T lymphocytes, plasma cells, and macrophages, with capillary endothelial proliferation and subsequent obliteration of small blood vessels. The cellular infiltration produces a T_H1 -type cytokine profile, consistent with the activation of macrophages. Phagocytosis of opsonized organisms by activated macrophages ultimately causes their destruction, resulting in spontaneous resolution of the chancre.

The generalized parenchymal, constitutional, mucosal, and cutaneous manifestations of secondary syphilis usually appear ~6–12 weeks after infection, although primary and secondary manifestations may occasionally overlap. In contrast, some patients may enter the latent stage without ever recognizing secondary lesions. The histopathologic features of secondary maculopapular skin lesions include hyperkeratosis of the epidermis, capillary proliferation with endothelial swelling in the superficial dermis, and—in the deeper dermis—perivascular infiltration by CD8+ T lymphocytes, CD4+ T lymphocytes, macrophages, and variable numbers of plasma cells. *T. pallidum* disseminates during the first days to weeks of infection, invading many tissues, including the CNS; cerebrospinal fluid (CSF) abnormalities can be detected in as many as 40% of patients during the secondary stage. Clinical hepatitis and immune complex-induced glomerulonephritis are rare, but recognized, manifestations of secondary syphilis. Generalized nontender lymphadenopathy is noted in 85% of patients with secondary syphilis. The paradoxical appearance of secondary manifestations, even after the development of an immune response that clears primary lesions, likely results from immune evasion due to antigenic variation of TprK surface antigens. Secondary lesions generally subside within

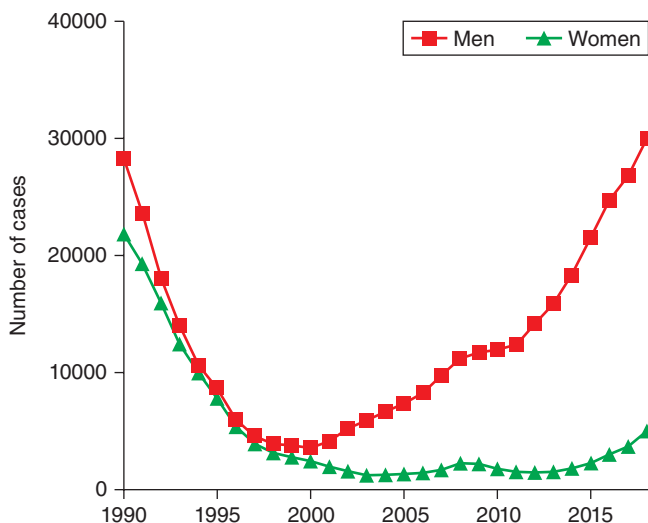


FIGURE 182-1 Primary and secondary syphilis in the United States, 1990–2018, by sex. (Data from the Centers for Disease Control and Prevention.)

1408 2–6 weeks, and the infection enters the latent stage, which is detectable only by serologic testing. In the preantibiotic era, up to 25% of untreated patients experienced at least one cutaneous relapse of secondary lesions, usually during the first year. Therefore, identification and examination of sexual contacts are most important for patients with syphilis of <1 year's duration.

In the preantibiotic era, about one-third of patients with untreated latent syphilis developed clinically apparent tertiary disease, the most common types being the gumma (a usually benign granulomatous lesion); cardiovascular syphilis (usually involving the vasa vasorum of the ascending aorta and resulting in aneurysm); and late symptomatic neurosyphilis (tabes dorsalis and paresis). In Western countries today, specific treatment for early and latent syphilis and coincidental therapy (i.e., therapy with antibiotics active against treponemes, but given for other conditions) have nearly eliminated tertiary syphilis. Asymptomatic CNS involvement, however, is still demonstrable in up to 40% of persons with early syphilis and 25% of patients with late latent syphilis, and modern cases of general paresis and tabes dorsalis are being reported from China. The factors that contribute to the development and progression of tertiary disease are unknown.

CLINICAL MANIFESTATIONS

■ PRIMARY SYPHILIS

The typical primary chancre usually begins as a single painless papule that rapidly erodes and becomes indurated, with a characteristic cartilaginous consistency on palpation of the edge and base of the ulcer. Multiple primary lesions are seen in a minority of patients. In heterosexual men, the chancre is usually located on the penis, where it is readily seen (Fig. 182-2; see also Fig. A1-17), but in MSM, it may also be found in the anal canal, rectum, or mouth. Oral sex has been identified as the source of infection in some MSM. In women, common primary sites are the cervix, vaginal wall, and labia, as well as anal canal and mouth. Consequently, primary syphilis goes unrecognized in women and MSM more often than in heterosexual men.

Atypical primary lesions are common, and may be multiple, small, or partially resolved. Therefore, syphilis should be considered in the evaluation of trivial or atypical dark-field-negative genital lesions. The

lesions that most commonly must be differentiated from those of primary syphilis include those caused by herpes simplex virus infection (Chap. 192), chancroid (Chap. 157), traumatic injury, and donovanosis (Chap. 173). Regional (usually inguinal) lymphadenopathy accompanies the primary syphilitic lesion, appearing within 1 week of lesion onset. The nodes are firm, nonsuppurative, and painless. Inguinal lymphadenopathy is bilateral and may occur with anal as well as with genital chancres. The chancre generally heals within 4–6 weeks (range, 2–12 weeks), but lymphadenopathy may persist for months.

■ SECONDARY SYPHILIS

The classical manifestations of the secondary stage include mucocutaneous or cutaneous lesions and generalized nontender lymphadenopathy. The healing primary chancre may still be present in ~15% of cases—more frequently in persons with concurrent HIV infection. The skin rash consists of macular, papular, papulosquamous, and occasionally pustular syphilides; often more than one form is present simultaneously. The eruption may be very subtle, and 25% of patients with a discernible rash may be unaware that they have dermatologic manifestations. Initial lesions are pale red or pink, nonpruritic, discrete macules distributed on the trunk and extremities; these macules progress to papular lesions that are distributed widely and that frequently involve the palms and soles (Fig. 182-3; see also Figs. A1-18 and A1-19). Rarely, severe necrotic lesions (*lues maligna*) may appear and are more commonly reported in HIV-infected individuals. Involvement of the hair follicles may result in patchy alopecia of the scalp hair, eyebrows, or beard in up to 5% of cases.

In warm, moist, intertriginous areas (commonly the perianal region, vulva, and scrotum), papules can enlarge to produce broad, moist, pink or gray-white, highly infectious lesions (*condylomata lata*; see Fig. A1-20) in 10% of patients with secondary syphilis. Superficial mucosal erosions (*mucous patches*) occur in 10–15% of patients and commonly involve the oral or genital mucosa (see Fig. A1-21). The typical mucous patch is a painless silver-gray erosion surrounded by a red periphery. *T. pallidum* DNA has been detected in oral mucosal swabs from persons with early syphilis, but who have no visible oral lesions. The implications of this finding for transmission are unclear but warrant further research.

Constitutional signs and symptoms that may accompany or precede secondary syphilis include sore throat (15–30%), fever (5–8%), weight loss (2–20%), malaise (25%), anorexia (2–10%), headache (10%), and meningismus (5%). *Acute meningitis* occurs in only 1–2% of cases, but CSF cell and protein concentrations are increased in up to 40% of early syphilis cases, and viable *T. pallidum* organisms have been recovered from CSF during primary and secondary syphilis in 30% of cases, sometimes without other CSF abnormalities. Persons with current or recent secondary syphilis may present with ocular or otic manifestations. Ocular findings include pupillary abnormalities and optic neuritis as well as the classic iritis or uveitis. The diagnosis of ocular syphilis is often considered in affected patients only after they fail to respond to topical steroid therapy. Anterior uveitis has been reported in 5–10% of patients with secondary syphilis, and *T. pallidum* has been demonstrated in aqueous humor from such patients. Permanent blindness may result without prompt diagnosis and treatment. Otic syphilis may present as sensorineural hearing loss, vertigo, or tinnitus. The recent publication of several reports of ocular and otic syphilis reminds clinicians to inquire about neurologic manifestations in all stages of syphilis infection. In a recent study, 7.9% of patients with syphilis, when asked, reported recent vision or hearing changes, and more than half of those had abnormal CSF or ophthalmologic findings consistent with syphilis.

Less often recognized complications of secondary syphilis include hepatitis, nephropathy, gastrointestinal involvement (hypertrophic gastritis, patchy proctitis, or a rectosigmoid mass—sometimes mistakenly assumed to be malignant), arthritis, and periostitis. Hepatic involvement is common in syphilis; although it is usually asymptomatic, up to 25% of patients may have abnormal liver function tests. Frank syphilitic hepatitis is rare. Renal involvement usually results from immune complex deposition and produces proteinuria associated with an acute nephrotic syndrome. Like those of primary syphilis, most



FIGURE 182-2 Primary syphilis with a firm, nontender chancre.



FIGURE 182-3 Secondary syphilis. *Left:* Maculopapular truncal eruption. *Middle:* Papules on the palms. *Right:* Papules on the soles. (Photos courtesy of Jill McKenzie and Christina Marra.)

manifestations of the secondary stage resolve spontaneously, usually within 1–6 months.

■ LATENT SYPHILIS

Positive serologic tests for syphilis, together with a normal CSF examination and the absence of clinical manifestations of syphilis, indicate a diagnosis of latent syphilis in an untreated person. The diagnosis may be made following routine serologic screening or may be suspected due to a history of primary or secondary lesions, a history of exposure to syphilis, or the delivery of an infant with congenital syphilis. A previous nonreactive serologic test or clear history of lesions or exposure may help to establish the duration of infection, which is an important factor in the selection of appropriate therapy. Early latent syphilis is limited to the first year after infection, whereas late latent syphilis is defined as that of ≥ 1 year's (or unknown) duration. The classical definition of early latent syphilis would include a person whose secondary rash has resolved, as well as a person whose chancre has healed but who has not yet developed secondary manifestations. The Centers for Disease Control and Prevention (CDC) have recently revised the case definitions for surveillance and reporting purposes to better reflect the recognition that some clinical manifestations may appear at several stages of infection. These definitions include the traditional primary and secondary stages, as well as “syphilis, early nonprimary nonsecondary,” describing infections of <12 months' duration, and “syphilis, unknown duration or late,” encompassing the previous late latent and late (tertiary) classifications. In this new scheme, neurologic, ocular, otic, and late clinical manifestations are reported separately in the context of their separate primary, secondary, early nonprimary nonsecondary, and unknown duration or late categories.

It was previously thought that untreated late latent syphilis had three possible outcomes: (1) persistent lifelong infection; (2) development of tertiary syphilis; or (3) spontaneous cure, with reversion of serologic tests to negative. Although progression to clinically evident late syphilis is very rare today, the occurrence of spontaneous microbiologic cure is in doubt.

Because *T. pallidum* continues to be present throughout untreated infection, it may seed the bloodstream intermittently during the latent stage, and a pregnant woman with latent syphilis may infect her fetus in utero. Moreover, syphilis has been transmitted through blood transfusion or organ donation from patients with latent syphilis.

■ REINFECTION SYPHILIS

A growing number of individuals, particularly MSM, are acquiring multiple episodes of syphilis, with important implications for clinical presentation and serologic testing. Although no national data are available, 32% of enrollees (mostly MSM) in a recent 18-year longitudinal study of CNS involvement were known to have had multiple episodes of syphilis. It is well recognized that, after treatment, persons with past syphilis are less likely to revert to nonreactive in the Venereal Disease Research Laboratory (VDRL)/rapid plasma reagin (RPR) than persons with first episode syphilis, and treponemal tests will remain

reactive. However, several recent studies also indicate that subsequent episodes of syphilis are more likely to be asymptomatic than initial episodes, less likely to have *T. pallidum* identified in blood or CSF, and less likely to have laboratory-defined neurosyphilis. These cases would be detectable only by serologic screening, reinforcing the utility of frequent screening in high-risk populations.

■ INVOLVEMENT OF THE CNS

Traditionally, neurosyphilis has been considered a late manifestation of syphilis, but this view is inaccurate. CNS syphilis represents a continuum encompassing early invasion (usually within the first weeks of infection), months to years

of asymptomatic involvement, and, in some cases, development of early or late neurologic manifestations. Early neurosyphilis includes asymptomatic or symptomatic meningitis and meningovascular syphilis; late neurosyphilis includes tabes dorsalis and general paresis.

Asymptomatic Neurosyphilis The diagnosis of asymptomatic neurosyphilis is made in patients who lack neurologic symptoms and signs but who have CSF abnormalities, including mononuclear pleocytosis, increased protein concentration, or reactivity in the CSF VDRL test. CSF abnormalities are demonstrated in up to 40% of cases of untreated primary or secondary syphilis and in 25% of cases of untreated latent syphilis. *T. pallidum* has been recovered by inoculation into rabbits of CSF from up to 30% of patients with primary or secondary syphilis but less frequently from patients with syphilis of >1 year's duration. The presence of *T. pallidum* in CSF is often associated with other CSF abnormalities, but organisms can be recovered from patients with otherwise normal CSF. Although the prognostic implications of these findings in early syphilis are uncertain, it may be appropriate to conclude that even patients with early syphilis who have CSF abnormalities do indeed have asymptomatic neurosyphilis and should be treated for neurosyphilis; such treatment is particularly important in patients with concurrent untreated HIV infection. Before the advent of penicillin, the risk of development of clinical neurosyphilis in untreated asymptomatic persons was roughly proportional to the intensity of CSF changes, with the overall cumulative probability of progression to clinical neurosyphilis ~20% in the first 10 years of infection but increasing with time. In several large studies, neurosyphilis was associated with an RPR titer of $\geq 1:32$, regardless of clinical stage or HIV infection status. While most experts agree that neurosyphilis is more common among persons with untreated HIV infection, the immune reconstitution seen with effective ART may have a protective effect against development of clinical neurosyphilis in HIV-infected persons with syphilis. Nonetheless, RPR titer $\geq 1:32$ is still associated with reactive CSF VDRL, even in persons taking effective ART. HIV-uninfected persons with untreated latent syphilis and normal CSF probably run a very low risk of subsequent neurosyphilis.

Symptomatic Neurosyphilis The major clinical categories of symptomatic neurosyphilis include early meningeal and meningovascular and late parenchymatous syphilis. The last category includes general paresis and tabes dorsalis. The onset of symptoms usually occurs <1 year after infection for meningeal syphilis, up to 10 years after infection for meningovascular syphilis, at ~20 years for general paresis, and at 25–30 years for tabes dorsalis. Neurosyphilis is more frequently symptomatic in patients co-infected with untreated HIV, particularly those with low CD4+ T lymphocyte counts. In addition, evidence suggests that syphilis infection worsens the cognitive impairment seen in HIV-infected persons and that this effect persists after treatment for syphilis.

Meningeal syphilis may present as headache, nausea, vomiting, neck stiffness, cranial nerve involvement, seizures, and changes in mental status. This condition may be concurrent with or may follow the

1410 secondary stage. Patients presenting with uveitis, iritis, or hearing loss often have meningeal syphilis, but these clinical findings can also be seen in patients with normal CSF.

Meningovascular syphilis reflects meningitis together with inflammatory vasculitis of small, medium, or large vessels. The most common presentation is a stroke syndrome involving the middle cerebral artery of a relatively young adult. However, unlike the usual thrombotic or embolic stroke syndrome of sudden onset, meningovascular syphilis often becomes manifest after a subacute encephalitic prodrome (with headaches, vertigo, insomnia, and psychological abnormalities), which is followed by a gradually progressive vascular syndrome.

The manifestations of general paresis reflect widespread late parenchymal damage and include abnormalities corresponding to the mnemonic *paresis*: personality, affect, reflexes (hyperactive), eye (e.g., Argyll Robertson pupils), sensorium (illusions, delusions, hallucinations), intellect (a decrease in recent memory and in the capacity for orientation, calculations, judgment, and insight), and speech. *Tabes dorsalis* is a late manifestation of syphilis that presents as symptoms and signs of demyelination of the posterior columns, dorsal roots, and dorsal root ganglia, including ataxia, foot drop, paresthesia, bladder disturbances, impotence, areflexia, and loss of positional, deep-pain, and temperature sensations. The small, irregular Argyll Robertson pupil, a feature of both *tabes dorsalis* and *paresis*, reacts to accommodation but not to light. *Optic atrophy* also occurs frequently in association with *tabes*.

■ OTHER MANIFESTATIONS OF LATE SYPHILIS

The slowly progressive inflammatory process leading to tertiary disease begins early during infection, although these manifestations may not become clinically apparent for years or decades. Early syphilitic aortitis first becomes evident soon after secondary lesions subside, and treponemes that trigger the development of gummas may have seeded the tissue years earlier.

Cardiovascular Syphilis Cardiovascular manifestations, usually appearing 10–40 years after infection, are attributable to endarteritis obliterans of the vasa vasorum, which provide the blood supply to large vessels; *T. pallidum* DNA has been detected by polymerase chain reaction (PCR) in aortic tissue. Cardiovascular involvement results in uncomplicated aortitis, aortic regurgitation, saccular aneurysm (usually of the ascending aorta), or coronary ostial stenosis. In the preantibiotic era, symptomatic cardiovascular complications developed in ~10% of persons with untreated late syphilis. Today, cardiovascular syphilis is rarely seen in the developed world.

Late Benign Syphilis (Gumma) Gummas are usually solitary lesions ranging from microscopic to several centimeters in diameter. Histologic examination shows a granulomatous inflammation, with a central area of necrosis due to endarteritis obliterans. *T. pallidum* has been detected by PCR in these lesions, and penicillin treatment results in rapid resolution, confirming the treponemal stimulus for the inflammation. Common sites include the skin and skeletal system; however, any organ (including the brain) may be involved. Gummas of the skin produce indolent, painless, indurated nodular or ulcerative lesions that may resemble other chronic granulomatous conditions. Skeletal gummas may affect any bone or cartilage. Upper respiratory gummas can lead to perforation of the nasal septum or palate.

■ CONGENITAL SYPHILIS

Transmission of *T. pallidum* across the placenta from a syphilitic woman to her fetus may occur at any stage of pregnancy, but fetal damage generally does not occur until after the fourth month of gestation when fetal immunologic competence begins to develop. This timing suggests that the pathogenesis of congenital syphilis, like that of adult syphilis, depends on the host immune response rather than on a direct toxic effect of *T. pallidum*. The risk of fetal infection during untreated early maternal syphilis is ~75–95%, decreasing to ~35% for maternal syphilis of >2 years' duration. Adequate treatment of the woman before the 16th week of pregnancy should prevent fetal damage, and treatment before the third trimester should adequately treat the infected fetus. Untreated maternal infection may result in a rate of fetal loss

of up to 40% with second-trimester spontaneous abortion, stillbirth, prematurity, and neonatal death. Among infants born alive, only fulminant congenital syphilis is clinically apparent at birth, and these babies have a very poor prognosis. The most common clinical problem is the healthy-appearing baby born to a mother with a positive serologic test.

Routine serologic testing for syphilis in early pregnancy is cost-effective in virtually all populations, even in areas with a low prenatal prevalence of syphilis. Low-tech point-of-care tests have been developed and widely implemented to facilitate antenatal testing in resource-poor settings. Globally, the past 10 years have seen a dramatic reduction in congenital syphilis in the face of relatively steady rates of maternal syphilis, showing the effectiveness of increased antenatal screening and treatment. Progress has been uneven, however, with major advances in Thailand, Cuba, several Baltic States, and India, but continuing high levels in Africa and China. Periodic lack of penicillin availability in low- and middle-income countries prevents treatment of seropositive women. Integration of programs to prevent congenital syphilis with programs to prevent maternal transmission of HIV would be highly cost-effective but is hampered by the restrictions placed on HIV-focused funds.

All pregnant women should be screened at their first antenatal visit. Where the prevalence of syphilis in women is high or when the patient is at high risk of reinfection, serologic testing should be repeated in the third trimester and at delivery. Neonatal congenital syphilis must be differentiated from other generalized congenital infections, including rubella, cytomegalovirus or herpes simplex virus infection, and toxoplasmosis, as well as from erythroblastosis fetalis.

Manifestations of congenital syphilis may appear early (within the first 2 years of life, often at 2–10 weeks of age) or late (after 2 years). The earliest manifestations of congenital syphilis include rhinitis, or “snuffles” (23%); mucocutaneous lesions (35–41%); bone changes (61%), including periostitis detectable by x-ray examination of long bones; hepatosplenomegaly (50%); lymphadenopathy (32%); anemia (34%); jaundice (30%); thrombocytopenia; and leukocytosis. CNS invasion by *T. pallidum* is detectable in 22% of infected neonates. Neonatal death is usually due to pulmonary hemorrhage, secondary bacterial infection, or severe hepatitis. Late congenital syphilis (untreated after 2 years of age) is subclinical in 60% of cases; the clinical spectrum in the remainder of cases may include interstitial keratitis (which occurs at 5–25 years of age), eighth-nerve deafness, and recurrent arthropathy. Neurosyphilis was documented in about one-quarter of untreated patients with late congenital syphilis in the preantibiotic era. Gummatous periostitis occurs at 5–20 years of age and, as in nonvenereal endemic syphilis, tends to cause destructive lesions of the palate and nasal septum. Classic stigmata include *Hutchinson's teeth* (centrally notched, widely spaced, peg-shaped upper central incisors), “mulberry” molars (sixth-year molars with multiple, poorly developed cusps), saddle nose, and saber shins.

LABORATORY EXAMINATIONS

■ DEMONSTRATION OF THE ORGANISM

Historically, dark-field microscopy and immunofluorescence antibody staining have been used to identify *T. pallidum* in moist lesions such as chancres or condylomata lata, but these tests are rarely available outside of research laboratories. Sensitive and specific PCR tests have been developed but are not commercially available, although a number of laboratories perform in-house validated PCR testing. The recent advances in cultivation of *T. pallidum* in a tissue culture system have not yet been implemented in clinical laboratories.

T. pallidum can be found in tissue by immunofluorescence or immunohistochemical methods using specific monoclonal or polyclonal antibodies to *T. pallidum*. Silver stains should be interpreted with caution because artifacts resembling *T. pallidum* are often seen. *T. pallidum* DNA has been detected by PCR in lesion swabs, tissue samples, blood, CSF, ocular fluid, urine, and oropharyngeal swabs.

■ SEROLOGIC TESTS FOR SYPHILIS

Treponemal and Lipoidal Tests There are two types of serologic tests for syphilis: lipoidal (so-called nontreponemal) and treponemal.

Both are reactive in persons with any treponemal infection, including syphilis, yaws, pinta, and endemic syphilis.

The most widely used lipoidal antibody tests are the RPR and VDRL tests, which measure IgG and IgM directed against a cardiolipin-lecithin-cholesterol antigen complex. The RPR test is easier to perform and uses unheated serum or plasma; it is the test of choice for rapid serologic diagnosis in a clinical setting. The VDRL test remains the standard for examining CSF and is superior to the RPR for this purpose. Either test is recommended for screening and for quantitation of serum antibody. The titer reflects disease activity, rising during early syphilis, often exceeding 1:32 in secondary syphilis, and declining slowly thereafter without therapy. After treatment for early syphilis, a persistent fall by fourfold or more (e.g., a decline from 1:32 to 1:8) is considered an adequate response. VDRL titers do not correspond directly to RPR titers, and sequential quantitative testing (as for response to therapy) must employ a single test. A reactive VDRL/RPR screening test must be confirmed by a treponemal test to rule out a biological false-positive reaction.

Treponemal tests measure antibodies to native or recombinant *T. pallidum* antigens and include the fluorescent treponemal antibody-absorbed (FTA-ABS) test and the *T. pallidum* particle agglutination (TPPA) test, both of which are more sensitive for primary syphilis than the lipoidal tests. When used to confirm reactive lipoidal test results, treponemal tests have a very high positive predictive value for diagnosis of syphilis.

Treponemal enzyme or chemiluminescence immunoassays (EIAs/CIAs), based largely on reactivity to recombinant antigens, are now widely used as screening tests by large laboratories. When these tests are used for screening, a high proportion of sera reactive by EIA/CIA are nonreactive by lipoidal tests. Such sera should be examined in the TPPA test, which includes different antigens and a different platform. If the TPPA test is nonreactive, the patient is unlikely to have syphilis; if it is reactive, the patient is likely to have current or past syphilis. Both lipoidal and treponemal tests may be nonreactive in early primary syphilis, although treponemal tests are slightly more sensitive (85–90%) during this stage than lipoidal tests (~80%). All tests are reactive during secondary syphilis. (Fewer than 1% of patients with high titers have a lipoidal test that is nonreactive or weakly reactive with undiluted serum but is reactive with diluted serum—the *prozone phenomenon*.) VDRL and RPR sensitivity and titers may decline in untreated persons with late latent syphilis, but treponemal tests remain reactive in late syphilis. After treatment for early syphilis, lipoidal test titers will generally decline or the tests will become nonreactive, whereas treponemal tests often remain reactive after therapy and are not helpful in determining the infection status of persons with past syphilis. There is some concern in the literature about persons in whom the lipoidal test titer fails to become nonreactive or remains reactive in low titer after treatment. The implications in such cases are unclear, but re-treatment rarely achieves the desired goal and is not recommended in the absence of clinical findings.

False-Positive Serologic Tests for Syphilis The lipid antigens of nontreponemal tests are similar to those found in human tissues, and these tests may be reactive (usually with titers $\leq 1:8$) in persons without treponemal infection, largely limited to persons with autoimmune conditions or injection drug use. Among patients being screened for syphilis because of risk factors, clinical suspicion, or history of exposure, ~1% of reactive tests are falsely positive. In a patient with a false-positive nontreponemal test, syphilis is excluded by a nonreactive treponemal test.

False-positive reactions may also occur with treponemal tests, particularly the EIA/CIA tests. Screening a low-prevalence population for syphilis with a treponemal test may result in true-positive reactions being outnumbered by false-positive reactions, leading to unnecessary treatment. Thus, screening with lipoidal tests is highly recommended.

■ EVALUATION FOR NEUROSYPHILIS

Involvement of the CNS is detected by examination of CSF for mononuclear pleocytosis (>5 white blood cells/ μL), increased protein concentration (>45 mg/dL), or CSF VDRL reactivity. Elevated CSF cell

counts and protein concentrations are not specific for neurosyphilis and may be confounded by HIV co-infection. Because CSF pleocytosis may also be due to HIV, some studies have suggested using a CSF white cell cutoff of 20 cells/ μL as diagnostic of neurosyphilis in HIV-infected patients with syphilis. The CSF VDRL test is highly specific and, when reactive, is considered diagnostic of neurosyphilis; however, this test is insensitive and may be nonreactive even in cases of symptomatic neurosyphilis. The RPR test should not be substituted for the VDRL test for CSF examination. The FTA-ABS test on CSF is reactive far more often than the CSF VDRL test in all stages of syphilis, but reactivity may reflect passive transfer of serum antibody into the CSF. A nonreactive FTA-ABS test on CSF, however, may be used to rule out asymptomatic neurosyphilis. Measuring CXCL13 in CSF has been demonstrated to distinguish between neurosyphilis and HIV-related CSF abnormalities.

All *T. pallidum*-infected patients with signs or symptoms consistent with neurologic disease (e.g., meningitis, hearing loss) or ophthalmic disease (e.g., uveitis, iritis) should have a CSF examination, regardless of disease stage. The appropriate management of asymptomatic persons is less clear. Lumbar puncture on all asymptomatic patients with untreated syphilis is impractical and unnecessary. Even at high doses, penicillin G benzathine fails to result in treponemicidal drug levels in CSF, and viable *T. pallidum* have been isolated from the CSF of patients (with and without HIV infection) after penicillin G benzathine treatment for early syphilis. Therefore, it is important to identify those persons at higher risk for having or developing neurosyphilis so that appropriate treatment may be given. Large-scale prospective studies have provided evidence-based guidelines for determining which syphilis patients may benefit most from CSF examination. Specifically, patients with RPR titers of $\geq 1:32$ are at higher risk of having neurosyphilis (11-fold and 6-fold higher in HIV-infected and HIV-uninfected persons, respectively), as are HIV-infected patients with CD4+ T-cell counts of $\leq 350/\mu\text{L}$. Persons with active tertiary syphilis and those in whom treatment failure is suspected should also have their CSF examined to determine appropriate therapy.

■ EVALUATION OF HIV-INFECTED PATIENTS FOR SYPHILIS

Because persons at highest risk for syphilis are also at increased risk for HIV infection, these two infections frequently coexist. There is evidence that syphilis and other genital ulcer diseases are important risk factors for acquisition and transmission of HIV infection. Some manifestations of syphilis may be altered in patients with concurrent untreated HIV infection, and multiple cases of neurologic relapse after standard therapy have been reported in these patients.

Persons with newly diagnosed HIV infection should be tested for syphilis; conversely, all patients with newly diagnosed syphilis should be tested for HIV infection. Some authorities, persuaded by reports of persistent *T. pallidum* in CSF of HIV-infected persons after standard therapy for early syphilis, recommend CSF examination for evidence of neurosyphilis for all co-infected patients, regardless of the stage of syphilis, with treatment for neurosyphilis if CSF abnormalities are found. Others, on the basis of their own clinical experience, think that standard therapy—without CSF examination—is sufficient for all cases of early syphilis in HIV-infected patients without neurologic signs or symptoms. As described above, RPR titer and CD4+ T-cell count can be used to identify patients at higher risk of neurosyphilis for lumbar puncture, although some cases of neurosyphilis will be missed even when these criteria are used. Serologic testing after treatment is important for all patients with syphilis, particularly for those also infected with HIV.

TREATMENT

Syphilis

TREATMENT OF ACQUIRED SYPHILIS

The CDC's 2015 guidelines for the treatment of syphilis are summarized in Table 182-1 and are discussed below. Penicillin G is the drug of choice for all stages of syphilis. *T. pallidum* is killed by very low concentrations of penicillin G, although a long period of

TABLE 182-1 Recommendations for the Treatment of Syphilis*

STAGE OF SYPHILIS	PATIENTS WITHOUT PENICILLIN ALLERGY	PATIENTS WITH CONFIRMED PENICILLIN ALLERGY
Primary, secondary, or early latent	CSF normal or not examined: Penicillin G benzathine (single dose of 2.4 mU IM) CSF abnormal: Treat as neurosyphilis.	CSF normal or not examined: Doxycycline (100 mg PO bid) or tetracycline HCl (500 mg PO qid) for 2 weeks CSF abnormal: Treat as neurosyphilis.
Late latent (or latent of unknown duration), cardiovascular, or benign tertiary	CSF normal or not examined: Penicillin G benzathine (2.4 mU IM weekly for 3 weeks) CSF abnormal: Treat as neurosyphilis.	CSF normal and patient not infected with HIV: Doxycycline (100 mg PO bid) or tetracycline HCl (500 mg PO qid) for 4 weeks CSF normal and patient infected with HIV: Desensitize and treat with penicillin if compliance cannot be assured. CSF abnormal: Treat as neurosyphilis.
Neurosyphilis (asymptomatic or symptomatic)	Aqueous crystalline penicillin G (18–24 mU/d IV, given as 3–4 mU q4h or continuous infusion) for 10–14 days or Aqueous procaine penicillin G (2.4 mU/d IM) plus oral probenecid (500 mg qid), both for 10–14 days	Desensitize and treat with penicillin.
Syphilis in pregnancy	According to stage	Desensitize and treat with penicillin.

*See text for indications for CSF examination.

Abbreviations: CSF, cerebrospinal fluid; mU, million units.

Source: Adapted from the 2015 Sexually Transmitted Diseases Treatment Guidelines from the Centers for Disease Control and Prevention. Available from <https://www.cdc.gov/std/tg2015/default.htm>.

exposure to penicillin is required because of the unusually slow rate of multiplication of the organism. The efficacy of penicillin against syphilis remains undiminished after 75 years of use, and there is no evidence of penicillin resistance in *T. pallidum*. Other antibiotics effective in syphilis include the tetracyclines and the cephalosporins. Aminoglycosides and spectinomycin inhibit *T. pallidum* only in very large doses, and the sulfonamides and most quinolones are inactive. Azithromycin showed significant promise as an effective oral agent against *T. pallidum*; however, strains harboring 23S rDNA mutations that confer macrolide resistance are widespread. Such strains represent >80–90% of recent isolates from large U.S., European, and Chinese cities, although the prevalence of resistant strains varies by geographic location. Routine treatment of syphilis with azithromycin is not recommended. Careful follow-up of any patient treated for syphilis with azithromycin must be assured.

Early Syphilis Patients and Their Contacts Penicillin G benzathine is the most widely used agent for the treatment of early syphilis (2.4 million units; Table 182-1), and for preventive treatment of individuals exposed to infectious syphilis within the previous 3 months. *The regimens recommended for prevention are the same as those recommended for early syphilis.* Penicillin G benzathine cures >95% of cases of early syphilis, although clinical relapse can follow treatment, particularly in patients with untreated HIV infection. Because the risk of neurologic relapse may be higher in HIV-infected patients, CSF examination is recommended for HIV-seropositive individuals with syphilis of any stage, particularly those with a serum RPR titer of $\geq 1:32$ or a CD4+ T-cell count of $\leq 350/\mu\text{L}$. Therapy appropriate for neurosyphilis should be given if there is any evidence of CNS infection.

Late Latent Syphilis or Syphilis of Unknown Duration If the CSF is normal or is not examined, the recommended treatment is penicillin G benzathine (7.2 million units total; Table 182-1). If CSF abnormalities are found, the patient should be treated for neurosyphilis.

Tertiary Syphilis CSF examination should be performed. If the CSF is normal, the recommended treatment is penicillin G benzathine (7.2 million units total; Table 182-1). If CSF is abnormal, the patient should be treated for neurosyphilis. The clinical response to treatment for benign tertiary syphilis is usually impressive, but responses in cardiovascular syphilis are not dramatic because aortic aneurysm and aortic regurgitation cannot be reversed by antibiotics.

Syphilis in Penicillin-Allergic Patients For penicillin-allergic patients with syphilis, a 2-week (early syphilis) or 4-week (late or late latent syphilis) course of therapy with doxycycline or tetracycline is recommended (Table 182-1). These regimens appear to be quite effective in early syphilis but have not been tested for late or late latent syphilis, and compliance may be problematic. Limited studies suggest that ceftriaxone (1 g/d, given IM or IV for 8–10 days) is effective for early syphilis. These nonpenicillin regimens have not been carefully evaluated in HIV-infected individuals and should be used with caution. If compliance and follow-up are not assured, penicillin-allergic HIV-infected persons with late latent or late syphilis should be desensitized and treated with penicillin.

Neurosyphilis Penicillin G benzathine, even at high doses, does not produce treponemicidal concentrations of penicillin G in CSF and should not be used for treatment of neurosyphilis. Asymptomatic neurosyphilis may relapse as symptomatic disease after treatment with benzathine penicillin, and the risk of relapse may be higher in HIV-infected patients. Both symptomatic and asymptomatic neurosyphilis should be treated with aqueous penicillin (Table 182-1). Administration of either IV aqueous crystalline penicillin G or of IM aqueous procaine penicillin G plus oral probenecid in recommended doses is thought to ensure treponemicidal concentrations of penicillin G in CSF. The clinical response to penicillin therapy for meningeal syphilis is dramatic, but treatment of neurosyphilis with existing parenchymal damage may only arrest disease progression. No data suggest that additional therapy (e.g., penicillin G benzathine for 3 weeks) would be beneficial after treatment for neurosyphilis.

The use of antibiotics other than penicillin G for the treatment of neurosyphilis has not been studied, although limited data suggest that 1–2 g/d of IV ceftriaxone for 10–14 days may be used. In patients with confirmed penicillin allergy, desensitization and treatment with penicillin are recommended.

Management of Syphilis in Pregnancy Every pregnant woman should undergo a lipoidal screening test at her first prenatal visit and, if at high risk of re-exposure, again in the third trimester and at delivery. In the untreated pregnant patient with presumed syphilis, expeditious treatment appropriate to the stage of the disease is essential. Patients should be warned of the risk of a Jarisch-Herxheimer reaction, which may be associated with mild premature contractions but rarely results in premature delivery.

Penicillin is the only recommended agent for the treatment of syphilis in pregnancy. If the patient has a documented penicillin allergy, desensitization and penicillin therapy should be undertaken according to the CDC's 2015 guidelines. After treatment, a quantitative nontreponemal test should be repeated monthly throughout pregnancy to assess therapeutic efficacy. Treated women whose antibody titers rise by fourfold or whose titers do not decrease by fourfold over a 3-month period should be re-treated.

EVALUATION AND MANAGEMENT OF CONGENITAL SYPHILIS

Whether or not they are infected, newborn infants of women with reactive serologic tests may themselves have reactive tests because of transplacental transfer of maternal IgG antibodies.

For asymptomatic infants born to women treated adequately with penicillin during the first or second trimester of pregnancy, monthly quantitative nontreponemal tests may be performed to monitor for appropriate reduction in antibody titers. Rising or persistent titers indicate infection, and the infant should be treated. Detection of neonatal IgM antibody is insensitive, and no commercially available test is currently recommended.

An infant should be treated at birth if (1) the treatment status of the seropositive mother is unknown; (2) the mother received inadequate or nonpenicillin therapy; (3) the mother received penicillin therapy in the third trimester; or (4) the infant may be difficult to follow. The CSF should be examined to obtain baseline values before treatment. Penicillin is the only recommended drug for the treatment of syphilis in infants. Specific recommendations for the treatment of infants and older children are included in the CDC's 2015 treatment guidelines.

JARISCH-HERXHEIMER REACTION

A dramatic although self-limited reaction consisting of fever, chills, myalgia, headache, tachycardia, increased respiratory rate, increased circulating neutrophil count, and vasodilation with mild hypotension may follow the initiation of treatment for syphilis. This reaction is thought to be a response to lipoproteins released by dying *T. pallidum* organisms. The Jarisch-Herxheimer reaction occurs in ~50% of patients with primary syphilis, 90% of those with secondary syphilis, and a lower proportion of persons with later-stage disease. Defervescence takes place within 12–24 h. In secondary syphilis, erythema and edema of cutaneous lesions may increase. Patients should be warned to expect such developments, which can be managed with symptom-based treatment. Steroid therapy is not required for this mild transient reaction.

FOLLOW-UP EVALUATION OF RESPONSES TO THERAPY

Efficacy of treatment should be assessed by clinical evaluation and monitoring of the quantitative VDRL or RPR titer for a fourfold decline (e.g., from 1:32 to 1:8). Patients with primary or secondary syphilis should be examined 6 and 12 months after treatment, and persons with latent or late syphilis at 6, 12, and 24 months. More frequent clinical and serologic examination (3, 6, 9, 12, and 24 months) is recommended for patients concurrently infected with HIV, regardless of the stage of syphilis.

After successful treatment of seropositive first-episode primary or secondary syphilis, the VDRL or RPR titer progressively declines; the test becomes nonreactive by 12 months in 40–75% of seropositive primary cases and in 20–40% of secondary cases. In patients with HIV infection or a history of prior syphilis, VDRL and RPR tests are less likely to become nonreactive. Rates of decline of serologic titers appear to be slower, and serologically defined treatment failures more common, among HIV-infected patients than among those without HIV co-infection; however, effective ART may reduce these differences. Re-treatment should be considered if serologic responses are not adequate or if clinical signs persist or recur. Because it is difficult to differentiate treatment failure from reinfection, the CSF should be examined, with treatment for neurosyphilis if CSF is abnormal and treatment for late latent syphilis if CSF is normal. A minority of patients treated for early syphilis may experience a one-dilution titer increase within 14 days after treatment; however, this early elevation does not significantly affect the serologic outcome at 6 months after treatment. Patients treated for late latent syphilis frequently have low initial VDRL or RPR titers and may not have a fourfold decline after therapy with penicillin. In such patients, re-treatment is not warranted unless the titer rises or signs and symptoms of syphilis appear. Because treponemal tests may remain reactive despite treatment for seropositive syphilis, these tests are not useful in following the response to therapy.

The activity of neurosyphilis (symptomatic or asymptomatic) correlates best with CSF pleocytosis, and this measure provides the most sensitive index of response to treatment. Repeat CSF examinations should be performed every 6 months until the cell count is

normal. An elevated CSF cell count falls to normal in 3–12 months in adequately treated HIV-uninfected patients. The persistence of mild pleocytosis in HIV-infected patients may be due to the presence of HIV in CSF; this scenario may be difficult to distinguish from treatment failure. Elevated levels of CSF protein fall more slowly, and the CSF VDRL titer declines gradually over several years. In patients treated for neurosyphilis, a fourfold reduction in serum RPR titer has been positively correlated with normalization of CSF abnormalities; this correlation is stronger in HIV-uninfected patients and in HIV-infected patients receiving effective ART.

IMMUNITY TO SYPHILIS

The rate of development of acquired resistance to *T. pallidum* after natural or experimental infection depends on both the size of the infecting inoculum and the duration of infection before treatment. Both humoral and cellular responses are important in the healing of early lesions. Cellular infiltration, predominantly by T lymphocytes and macrophages, produces an interferon γ -dominated cytokine milieu and results in the clearance of organisms by activated macrophages. Specific antibodies to surface antigens enhance phagocytosis. Antigenic variation of the TprK protein contributes to development of subsequent stages of syphilis, persistence of infection, and susceptibility to reinfection with another strain. Comparative genomic studies have revealed genes with sequence variations among *T. pallidum* strains, leading to development of molecular typing methods used to examine syphilis outbreaks. Recent work has demonstrated that immunization with the outer-membrane protein Tp0751 significantly reduces dissemination of *T. pallidum* during syphilis infection in an animal model. Vaccine studies with this and other antigens are underway.

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Endemic Treponematoses

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The endemic treponematoses are chronic diseases that are transmitted by direct contact, usually during childhood and, like syphilis, can cause severe late manifestations years after initial infection. These diseases are caused by very close relatives of *Treponema pallidum* subspecies *pallidum*, the etiologic agent of venereal syphilis (Chap. 182). Yaws, pinta, and endemic syphilis (bejel) have traditionally been distinguished from venereal syphilis by mode of transmission, age of acquisition, geographic distribution, and clinical features; however, there is overlap for each of these factors. Our “knowledge” about these infections is based on observations by health care workers who have visited endemic areas during the past 70 years. Except for the ongoing programs of mass drug administration (MDA) for yaws eradication promoted by the World Health Organization (WHO), virtually no well-designed studies of the natural history, diagnosis, or treatment of